

HANNOVER, GermanyWMO: **103380**Lat: **52.4644N**Lon: **9.6781E**Elev: **195**StdP: **14.59**Time Zone: **1.00 (EUC)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.8	19.7	10.1	9.3	18.9	14.3	11.4	22.3	28.2	45.7	25.4	45.0	8.4	90	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	86.3	67.5	82.5	66.2	78.9	64.6	69.3	81.9	67.7	79.1	66.2	76.2	9.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	96.0	73.2	64.2	90.5	71.4	62.5	85.3	70.0	33.4	81.2	32.1	79.6	30.9	76.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	19.9	17.9		10.4	93.3	6.2	2.9	5.9	95.4	2.3	97.1	-1.2	98.7	-5.7	100.8
				9.8	73.3	6.3	1.5	5.3	74.4	1.6	75.3	-1.9	76.1	-6.4	77.2
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	35.1	35.9	41.5	50.1	55.9	62.3	66.3	64.7	59.3	51.6	43.7	38.8
	DBStd	12.85	8.39	7.50	7.58	6.86	7.09	6.16	6.06	5.40	5.43	6.53	6.91	7.96
	HDD50	1863	463	396	274	84	22	1	0	0	1	61	210	351
	HDD65	5516	927	815	728	447	293	124	56	71	188	415	640	812
	CDD50	2050	1	0	11	86	206	369	506	456	280	111	20	4
	CDD65	231	0	0	0	0	13	43	97	62	16	0	0	0
	CDH74	2191	0	0	0	16	140	413	944	525	145	8	0	0
	CDH80	645	0	0	0	0	25	101	332	159	28	0	0	0
(14) Wind	WSAvg	8.9	10.8	9.8	9.8	8.6	8.4	7.9	7.9	7.6	7.7	8.3	9.2	10.3
(15) Precipitation	PrecAvg	25.3	2.0	1.5	1.8	1.7	2.3	2.5	2.6	2.6	2.0	1.9	2.1	2.3
	PrecMax	36.9	5.9	4.4	5.2	4.2	6.7	6.1	6.2	6.8	4.7	5.4	4.5	6.3
	PrecMin	17.0	0.1	0.2	0.4	0.3	0.3	0.8	0.9	0.4	0.2	0.4	0.1	0.2
	PrecStd	4.3	1.1	0.9	1.0	0.9	1.4	1.2	1.2	1.2	1.0	1.0	1.0	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.0	66.1	76.6	83.9	87.7	93.0	91.1	84.1	73.5	62.4	57.1
		MCWB	50.5	45.9	52.5	59.2	67.0	68.0	69.3	68.9	65.3	60.8	56.6	52.5
	2%	DB	51.9	51.5	60.7	71.7	77.4	82.7	87.9	84.3	78.5	67.9	58.8	53.9
		MCWB	48.5	46.9	50.6	55.9	62.9	66.4	67.1	67.4	63.3	60.1	54.4	50.6
	5%	DB	48.6	48.5	55.7	67.9	73.6	78.8	82.8	79.2	73.5	64.3	55.7	51.6
		MCWB	46.4	45.1	48.6	55.2	59.4	65.0	66.1	64.8	62.4	58.1	52.1	48.5
	10%	DB	46.4	46.3	52.2	63.0	69.8	75.0	79.0	75.4	69.6	61.0	53.7	48.5
		MCWB	44.4	43.1	46.6	53.4	58.9	63.0	65.0	63.0	61.1	56.6	50.0	46.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	49.6	55.1	60.5	68.0	70.9	72.0	70.8	67.5	62.3	58.1	53.1
		MCDB	54.4	50.9	62.7	72.7	81.2	82.7	83.4	84.0	78.0	69.3	61.6	56.2
	2%	WB	49.3	47.3	52.1	58.0	64.8	68.1	69.5	68.5	65.4	60.6	55.0	51.3
		MCDB	51.8	50.4	58.4	68.0	73.8	78.6	82.2	80.9	75.3	67.3	57.9	53.7
	5%	WB	46.6	45.1	49.9	56.0	62.1	66.0	68.0	66.7	63.5	58.8	52.7	48.9
		MCDB	48.7	47.9	54.7	65.2	71.0	76.0	80.2	76.8	71.5	63.6	56.1	51.2
	10%	WB	44.1	42.8	47.5	54.3	59.4	64.2	66.4	64.9	61.6	56.9	50.6	46.3
		MCDB	45.7	45.6	52.2	62.9	67.5	73.1	76.8	73.3	68.3	61.1	53.4	48.6
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	9.7	13.7	18.2	17.9	18.0	17.9	17.2	16.0	12.6	8.7	7.9
		MCDBR	9.1	13.1	20.5	26.0	24.4	24.6	25.3	24.2	23.2	16.7	10.9	9.5
		MCWBR	8.1	9.5	12.8	13.6	12.3	11.7	9.9	10.6	11.4	10.3	8.1	8.0
		MCDBR	8.7	12.1	17.8	22.5	21.6	22.0	22.0	20.6	19.7	15.2	10.2	9.1
	5% WB	MCWBR	8.0	9.6	11.7	12.6	12.1	11.3	9.5	10.1	10.5	9.9	8.0	8.1
(40) Clear-Sky Solar Irradiance	taub		0.318	0.339	0.373	0.411	0.412	0.410	0.425	0.418	0.386	0.356	0.337	0.313
	taud		2.372	2.357	2.279	2.218	2.252	2.276	2.243	2.278	2.360	2.426	2.401	2.363
	Ebn at Noon		213	244	256	260	266	268	260	255	250	234	204	192
	Edh at Noon		19	25	33	40	41	40	41	37	31	24	18	16
(44) All-Sky Solar Radiation	RadAvg		202	422	758	1250	1531	1639	1564	1314	959	549	247	158
	RadStd		20	60	84	149	166	126	208	101	117	53	27	15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-223	N/A	N/A